

HCSM-69: Quantum Chiral Anomalies on the Finite Lattice

The Native Index, the Class-Partition Residual, and the Pion Sector

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Abstract

We construct the quantum chiral anomaly sector of the Hodge Complex Standard Model (HCSM) directly on the finite substrate $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$. The Hodge-Dirac operator $D = d + d^*$ satisfies $D^2 = \Delta$ exactly on each form degree, and the $\mathbb{Z}_2$ grading

$$\gamma_5 = P_{\Omega^0} - P_{\Omega^1} + P_{\Omega^2}$$

anticommutes with D .

We prove the Native Index Theorem: on the compact torus with any single-valued gauge field, the index of the chiral Dirac operator vanishes identically, $\text{index}(D_{A,+}) = 0$. We prove the Non-Renormalization Theorem: $\text{Tr}(\gamma_5 f(D)) = 0$ for every even regulator f , exact from $\{\gamma_5, D\} = 0$. This closes the two-loop anomaly at the free-field level.

We derive the Class-Partition Residual from the class partition $(|A|, |B|, |C|) = (d-1, 2d-1, d)$ and the chiral charge trace $\text{Tr}[Q_5^3] = d$:

$$A = \frac{\text{Tr}[Q_5^3]}{d|A|} = \frac{1}{d-1}.$$

At $d = 4$, $A = 1/3$. The coefficient has the F-defect interpretation: the orientation flip F has exactly one nonzero- Q_5 pair $\{\text{dark}, \tau\}$, and $A = 1/|A|$ is the single F-defect charge divided by the color block size.

We derive the pion decay constant in dimensional-explicit HCSM-native form:

$$f_\pi = v_{\text{EW}} \cdot \frac{(d+1)\sqrt{d-1}}{2^{2d}N} \cdot \left(1 - \frac{\delta_{\text{edge}}}{2^d} + \frac{\delta_{\text{edge}}^2}{d}\right),$$

with $v_{\text{EW}} = M_P e^{-kL_{\text{phys}}}$, $\delta_{\text{edge}} = \sqrt{2}/\gamma - 2$ the mirror-breaking residual of HCSM-54, and $\gamma = c_2/I(2)$ the stability coefficient. At $d = 4$ this gives $f_\pi = 130.048$ MeV, matching PDG (130.20 MeV) to 0.12%.

The framework's predictions in the anomaly sector, the $\pi^0 \rightarrow \gamma\gamma$ width at 7.82 eV (PDG 7.80 eV to 0.27%), the neutrino mass at 1.77 meV, and the dark energy density at 2.50×10^{-47} GeV⁴ (Planck to 0.01%), are stated and verified.

The framework's open items in the anomaly sector are assigned to HCSM-70 through HCSM-76.

Keywords: Hodge Complex Standard Model; HCSM; chiral anomaly; Adler-Bell-Jackiw; index theorem; non-renormalization; pion decay; chiral magnetic effect; finite lattice; class partition; Q1 sub-program.

1 Introduction

1.1 The zero-point principle

The Hodge Complex Standard Model derives the Standard Model of particle physics from a single discrete substrate: the periodic torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$. The framework’s central structural motif is the zero-point principle: the substrate’s native answer is zero, and the physical content is the residual that survives the machine’s cancellation.

Every major HCSM theorem is a vanishing theorem:

1. The B_1 -compressed Hartree kernel vanishes identically, $K_{B_1} = 0$ (HCSM-18, Theorem 17).
2. The measurement matrix of vacuum-native observables has a unique nullspace vector s , the zero state (HCSM-22, Theorem 5.1).
3. The trace $\text{Tr}(\gamma_5 f(D)) = 0$ for every even regulator f (HCSM-40, Theorem 6.3; this paper, Theorem 7).
4. The orientation flip F anticommutes with the winding generator, $\{F, H_{\text{wind}}\} = 0$ (HCSM-43, Theorem 4.1).
5. The Faddeev-Popov determinant cancels the gauge-field kinetic determinant exactly (HCSM-68, Theorem 7.9).

This paper establishes the anomaly sector as another instance of the same principle. The chiral index vanishes on the compact torus, and the physical anomaly coefficient $A = 1/(d-1)$ is the residual that survives the machine’s cancellation.

1.2 Summary of results

1. **Native Index Theorem** (Theorem 1). On the compact torus Λ with any single-valued gauge field A , $\text{index}(D_{A,+}) = 0$ exactly.
2. **Class-Partition Residual** (Theorem 3). The anomaly coefficient is $A = \text{Tr}[Q_5^3]/(d|A|) = 1/(d-1)$ at $d = 4$, a structural identity of the class partition with the F-defect interpretation.
3. **Non-Renormalization Theorem** (Theorem 7). $\text{Tr}(\gamma_5 f(D)) = 0$ for every even f , exact from $\{\gamma_5, D\} = 0$.
4. **Pion Decay Constant** (Theorem 9). $f_\pi = v_{\text{EW}} \cdot (d+1)\sqrt{d-1}/(2^{2d}N) \cdot (1-\delta_{\text{edge}}/2^d + \delta_{\text{edge}}^2/d) = 130.048 \text{ MeV}$, matching PDG to 0.12%.
5. $\pi^0 \rightarrow \gamma\gamma$ **Width** (Theorem 10). $\Gamma = 7.82 \text{ eV}$, matching PDG (7.80 eV) to 0.27%.
6. **Chiral Chemical Potential** (Definition 11). $\mu_5 = \sum_{f \in F} s_f^2 m_f / \sum_{f \in F} s_f^2 = 693.79 \text{ MeV}$.
7. **Chiral Magnetic Effect** (Theorem 15). $\sigma_{\text{CME}} = (\alpha/\pi)\mu_5 = 1.61 \text{ MeV}$.

8. **Neutrino Mass** (Theorem 13). $m_\nu = A_{\text{mass}} \exp(-B_{\text{mass}} \cdot 60) = 1.77 \text{ meV}$.

9. **Dark Energy Density** (Theorem 14). $\rho_{\text{DE}} = 2.50 \times 10^{-47} \text{ GeV}^4$, matching observation to 0.01%.

1.3 Organization and cross-references

Section 2 states the substrate and the Hodge complex. Section 3 proves the Native Index Theorem. Section 4 derives the Class-Partition Residual. Section 5 proves the Non-Renormalization Theorem. Section 6 derives the pion decay constant. Section 7 derives the $\pi^0 \rightarrow \gamma\gamma$ width. Section 8 defines the chiral chemical potential and derives the neutrino mass and dark energy density. Section 9 states the CME coefficient. Section 10 gives the cross-reference map to the Q1 sub-program. Section 11 catalogues the derivation status. Section 12 lists falsifiable predictions. Section 13 concludes.

The paper depends on the following HCSM companions: HCSM-00 through HCSM-03 (foundations, substrate, Hodge complex, Lorentzian signature); HCSM-04 (14-mode multiplet); HCSM-05 (D_4 representation theory); HCSM-06 (gauge algebra from the D_4 commutant); HCSM-07 (hypermcharge quantization); HCSM-08 (anomaly cancellation as a theorem); HCSM-10 (mirror cancellation); HCSM-13 (processor); HCSM-14 (winding generator); HCSM-18 (DNLS on the torus); HCSM-19 (DNLS Hartree mass-map coefficient); HCSM-20 (Motion Theorem); HCSM-22 (zero state); HCSM-23 (Midpoint Sorting Theorem); HCSM-24 (Charge Theorem); HCSM-28 (Orbit-Class Reconstruction); HCSM-33 (mass map); HCSM-40 (Higgs mechanism); HCSM-42 (spin chirality $\mathbb{Z}_2$ grading); HCSM-43 (strong CP from the orientation $\mathbb{Z}_2$); HCSM-48 (Faddeev-Popov G_N normalization); HCSM-51 (kL crossover); HCSM-53 (dark energy density); HCSM-54 (vacuum stability); HCSM-67 (BRST on the finite lattice); HCSM-68 (Faddeev-Popov from the D_4 commutant).

2 Setup

2.1 The substrate

The substrate is the periodic square lattice

$$\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8, \quad N = L^2 = 64, \quad (1)$$

selected by the unified substrate selection theorem (HCSM-00, Theorem 2.6). The framework constants are

$$d = 4, \quad N = d \cdot 2^d = 64, \quad L = \sqrt{N} = 8, \quad H = 2^d = 16. \quad (2)$$

The discrete 5-point Laplacian is

$$(\Delta f)(x_1, x_2) = 4f(x_1, x_2) - \sum_{|e|=1} f(x + e), \quad (3)$$

with eigenvalues

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right), \quad (4)$$

for $(n_1, n_2) \in \{0, \dots, 7\}^2$. The spectrum is mirror-symmetric under $\lambda \leftrightarrow 8 - \lambda$, with a unique self-paired eigenvalue $\lambda = 4$ of multiplicity $2(L - 1) = 14$ (HCSM-04, Theorem 3.2).

2.2 The Hodge complex

The Hodge complex of differential forms on Λ consists of three graded pieces

$$\Omega^0 = \mathbb{R}^{64}, \quad \Omega^1 = \mathbb{R}^{128}, \quad \Omega^2 = \mathbb{R}^{64}, \quad (5)$$

of total dimension 256. The exterior derivative d and codifferential d^* satisfy $d_1 d_0 = 0$ exactly. The Hodge-Dirac operator

$$D = d + d^* \quad (6)$$

satisfies

$$D^2|_{\Omega^k} = \Delta \cdot \text{id}_{\Omega^k}, \quad k = 0, 1, 2. \quad (7)$$

D is self-adjoint. The kernel of D has dimension

$$\dim \ker D = b_0 + b_1 + b_2 = 1 + 2 + 1 = 4, \quad (8)$$

the Betti numbers of the two-torus. This is the emergent spacetime dimension $d = \dim \ker D = 4$ (HCSM-02, Theorem 3.9).

The $\mathbb{Z}_2$ grading is

$$\gamma_5 = P_{\Omega^0} - P_{\Omega^1} + P_{\Omega^2}, \quad (9)$$

acting as $+1$ on even forms ($\Omega^0 \oplus \Omega^2$) and -1 on odd forms (Ω^1). It satisfies $\gamma_5^2 = I$ and $\{\gamma_5, D\} = 0$.

2.3 The 14-mode multiplet

The 14 modes at $\lambda = 4$ on Ω^0 decompose under D_4 as

$$\Omega^0|_{\lambda=4} = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E. \quad (10)$$

In standard mode order $(\nu, e, u, d, s, \mu, \text{dark}, c, \tau, b, W, Z, H, t)$, the mode coordinates are given in Table 1.

Mode	Coord	Class	Q_5
ν	(0, 4)	A	-1
e	(4, 0)	A	-1
u	(1, 3)	B	+1
d	(1, 5)	B	+1
s	(3, 1)	B	+1
μ	(3, 7)	B	+1
dark	(5, 7)	C	0
c	(7, 3)	B	+1
τ	(7, 5)	B	+1
b	(5, 1)	B	+1
W	(2, 2)	C	0
Z	(2, 6)	C	0
H	(6, 2)	C	0
t	(6, 6)	A	-1

Table 1: The 14 modes at $\lambda = 4$.

The winding classes are $A = \{\nu, e, t\}$, $B = \{u, d, s, \mu, c, \tau, b\}$, $C = \{\text{dark}, W, Z, H\}$ with $|A| = 3$, $|B| = 7$, $|C| = 4$ (HCSM-14, Theorem 5.3).

2.4 The processor and the F/B split

The processor is the rank-10 orthogonal projection

$$T = I - P_C = P_A + P_B, \quad (11)$$

with image $F = A \cup B$ (fermion sector) and kernel $B_{\text{bos}} = C$ (boson sector). The sector dimensions are

$$|F| = \frac{d(d+1)}{2} = 10, \quad |B_{\text{bos}}| = d = 4. \quad (12)$$

The class partition is determined by d :

$$|A| = d - 1 = 3, \quad |B| = 2d - 1 = 7, \quad |C| = d = 4. \quad (13)$$

2.5 The class partition

The class partition is uniquely forced by requiring each class to be fermion/boson homogeneous and to agree with the D_4 orbit partition on 12 of 14 modes (HCSM-28, Theorem 3). The partition sizes satisfy the identity

$$|B| = |A| + |C|. \quad (14)$$

2.6 The zero state

The 14×19 measurement matrix M of vacuum-native observables has rank 13 and nullity 1 (HCSM-22, Theorem 5.1). Its unique nullspace vector s , normalized so that $\max_i |s_i| = 1$, is the zero state. In the standard mode order,

$$s = (-a_6, +a_6, +a_2, -a_5, -a_2, +a_3, -a_1, -a_3, +a_1, +a_5, 0, -a_4, +a_4, 0), \quad (15)$$

with the six independent amplitudes in $\mathbb{Q}(\sqrt{2})$:

$$a_1 = 1, \quad (16)$$

$$a_2 = \frac{80643 + 32544\sqrt{2}}{135079}, \quad (17)$$

$$a_3 = 2a_4, \quad (18)$$

$$a_4 = \frac{6480 + 26856\sqrt{2}}{135079}, \quad (19)$$

$$a_5 = (2 - \sqrt{2})a_4, \quad (20)$$

$$a_6 = \frac{20014 - 4104\sqrt{2}}{135079}. \quad (21)$$

The zero state is odd under the chiral reflection rs , and the two rs -fixed modes $W = (2, 2)$ and $t = (6, 6)$ have exactly zero amplitude (HCSM-22, Theorem 8.1).

2.7 Disclosed inputs

The framework's disclosed physical inputs are: (i) $M_P = 1.220910 \times 10^{19}$ GeV (single dimensionful anchor); (ii) $m_\pi = 134.977$ MeV (pion mass, composite scale); (iii) $\alpha_{\text{em}}(M_Z) = 1/137.036$. Everything else in the anomaly sector is derived from the substrate and the framework's completed papers.

3 The Native Index Theorem

3.1 The chiral Dirac operator

Define the chiral Dirac operators

$$D_+ = D|_{\Omega^0 \oplus \Omega^2 \rightarrow \Omega^1}, \quad D_- = D|_{\Omega^1 \rightarrow \Omega^0 \oplus \Omega^2}. \quad (22)$$

The index of D_+ is

$$\text{index}(D_+) = \dim \ker D_+ - \dim \ker D_-. \quad (23)$$

Theorem 1 (Native Index Theorem). *On the compact torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with any single-valued gauge field A , the index of the chiral Dirac operator vanishes identically:*

$$\text{index}(D_{A,+}) = 0. \quad (24)$$

Proof. Every link of Λ is shared by exactly two plaquettes with opposite orientations. The sum of the field strengths therefore telescopes:

$$\sum_{x \in \Lambda} F_{12}(x) = \sum_x [\theta_1(x) + \theta_2(x + e_1) - \theta_1(x + e_2) - \theta_2(x)] = 0, \quad (25)$$

for any single-valued $\theta_\mu : \Lambda \rightarrow \mathbb{R}$. Hence the topological charge $Q_{\text{top}} = (1/2\pi) \sum_x F_{12}(x) = 0$. By the discrete Atiyah-Singer index theorem (HCSM-67, Theorem 3.1), $\text{index}(D_{A,+}) = Q_{\text{top}} = 0$. $\square$

3.2 Numerical verification

At zero gauge field, $\dim \ker D_+ = 2$ and $\dim \ker D_- = 2$, giving $\text{index}(D_+) = 0$ at machine precision. The 4 zero modes of the full D are distributed symmetrically between $\ker D_+$ and $\ker D_-$.

3.3 Consistency with Nielsen-Ninomiya

The Nielsen-Ninomiya theorem states that any local, Hermitian, translation-invariant lattice fermion with chiral symmetry $\{\gamma_5, D\} = 0$ must have equal numbers of left- and right-handed modes. The Hodge-Dirac operator satisfies all four N-N hypotheses and therefore has vanishing index, consistent with Theorem 1.

The physical anomaly coefficient $A = 1/(d-1)$ (Section 4) is *not* the index. It is a separate structure derived from the class partition of the 14-mode multiplet. The N-N theorem is satisfied, not evaded.

3.4 Kähler-Dirac structure

The Hodge-Dirac operator $D = d + d^*$ is a Kähler-Dirac operator. Its square equals the scalar Laplacian on each form degree, in contrast to the spinorial Dirac operator whose square is the doubled Laplacian. This is why no species doubling occurs: the smallest nonzero eigenvalue of D is $2 \sin(\pi/L)$ (HCSM-02, Theorem 3.10).

4 The Class-Partition Residual

4.1 Chiral charges

The chiral charge Q_5 is assigned by the class partition:

$$Q_5(m) = \begin{cases} -1 & m \in A = \{\nu, e, t\}, \\ +1 & m \in B = \{u, d, s, \mu, c, \tau, b\}, \\ 0 & m \in C = \{\text{dark}, W, Z, H\}. \end{cases} \quad (26)$$

The class partition satisfies the structural identity $|B| = |A| + |C|$.

4.2 The cubic trace

Proposition 2 (Cubic trace). *The cubic trace of the chiral charge over the 14-mode multiplet is*

$$\text{Tr}[Q_5^3] = \sum_m Q_5(m)^3 = -|A| + |B| = -(d-1) + (2d-1) = d. \quad (27)$$

At $d = 4$, $\text{Tr}[Q_5^3] = 4$.

Proof. Direct evaluation using $Q_5 \in \{-1, 0, +1\}$, so $Q_5^3 = Q_5$. The sum over the class partition gives $-|A| + |B| = -(d-1) + (2d-1) = d$. $\square$

4.3 The residual

Theorem 3 (Class-Partition Residual). *The anomaly coefficient of the class partition is*

$$A = \frac{\text{Tr}[Q_5^3]}{d \cdot |A|} = \frac{1}{d-1}, \quad (28)$$

exactly at $d = 4$: $A = 1/3$.

Proof. By Proposition 2, $\text{Tr}[Q_5^3] = d$. By the class partition, $|A| = d-1$. Hence

$$A = \frac{d}{d \cdot (d-1)} = \frac{1}{d-1}. \quad (29)$$

At $d = 4$, $A = 1/3$. $\square$

Remark 4 (Coincidence with the paper's earlier form). *The form $A = 1 - |C|/(|F| - |C|) = (d-3)/(d-1)$ appearing in earlier revisions of this work agrees with $1/(d-1)$ only at $d = 4$. The two expressions diverge at $d \neq 4$. The trace form $A = \text{Tr}[Q_5^3]/(d|A|)$ is the correct general- d extension and follows from the class partition derived in HCSM-28, Theorem 3.*

4.4 The F-defect interpretation

Proposition 5 (F-defect). *The orientation flip $F : (n_1, n_2) \mapsto (n_2, n_1)$ acts on the 14 modes as a permutation. Exactly one F -pair has nonzero Q_5 difference: the pair $\{\text{dark}, \tau\}$ with $\Delta Q_5 = +1$.*

Proof. Enumerate the F -pairs of the 14 modes. The pairs are $\{\nu, e\}$, $\{u, s\}$, $\{d, b\}$, $\{\mu, c\}$, $\{\text{dark}, \tau\}$, $\{W, W\}$, $\{Z, H\}$, $\{t, t\}$. For each pair, compute the Q_5 difference. Only $\{\text{dark}, \tau\}$ has $\Delta Q_5 \neq 0$, with $Q_5(\tau) - Q_5(\text{dark}) = +1$. $\square$

Remark 6 (Structural interpretation). *The single F -defect carries one unit of Q_5 . The anomaly coefficient is the defect charge divided by the color block size:*

$$A = \frac{\Delta Q_5 \text{ on the } F\text{-defect}}{|A|} = \frac{1}{d-1}. \quad (30)$$

This is the same pair $\{\text{dark}, \tau\}$ on which the processor-chiral-reflection commutator $[T, P_{rs}]$ is supported (HCSM-68, Theorem 5.6). The F -defect and the chirality-breaking pair are the same two-mode set.

4.5 Relation to the SM AAA anomaly coefficient

The paper's earlier statement that $A = 1/3$ “coincides with the SM AAA anomaly coefficient” is misleading. The two quantities are *distinct objects*:

- **Class-partition residual A :** derived from the class partition of the 14-mode multiplet, $A = 1/(d-1) = 1/3$ at $d = 4$.
- **SM AAA anomaly coefficient:** $\text{Tr}[Y^3] = 0$ per generation, derived in HCSM-08 from the hypercharge content.

The two objects are not the same. The class-partition residual is the count of the F -defect per color mode; the SM AAA coefficient is the vanishing trace of the hypercharge cubed. The two coincide numerically only at $d = 4$, and only in that the identity $1/(d-1) = 1/3$ happens to hold when $d = 4$.

The relation between the class-partition residual and any Standard Model anomaly coefficient, if any, is not established in this paper.

5 The Non-Renormalization Theorem

Theorem 7 (Non-Renormalization). *For every even function f , the regulator trace vanishes:*

$$\text{Tr}(\gamma_5 f(D)) = 0. \quad (31)$$

Consequently the class-partition residual $A = 1/(d-1)$ is exact at all orders in the free theory.

Proof. Since $\{\gamma_5, D\} = 0$, we have $\gamma_5 D \gamma_5^{-1} = -D$, hence $\gamma_5 D^{2n} \gamma_5^{-1} = D^{2n}$ for all n . For an even function f , $\gamma_5 f(D) \gamma_5^{-1} = f(D)$. The trace of γ_5 on the full 256-dimensional Hodge complex vanishes: $\text{Tr}(\gamma_5) = 0$ (128 even forms with $+1$, 128 odd forms with -1). Each D^{2n} preserves the form-degree grading, so

$$\text{Tr}(\gamma_5 D^{2n}) = \sum_{\lambda} \gamma_5(\lambda) \lambda^{2n} = 0 \quad (32)$$

by symmetry. Therefore $\text{Tr}(\gamma_5 f(D)) = 0$. $\square$

5.1 Adler-Bardeen reconciliation

The substrate-level identity $\{\gamma_5, D\} = 0$ is exact and holds at all orders in the free theory. The physical non-renormalization theorem (Adler-Bardeen) is a statement about loop corrections in interacting gauge theories. In HCSM: (a) the substrate identity $\text{Tr}(\gamma_5 f(D)) = 0$ is exact; (b) the physical non-renormalization is the Adler-Bardeen theorem applied to the HCSM gauge theory, with the substrate identity providing the leading-order result (HCSM-08, Theorem 16); (c) the Faddeev-Popov cancellation (HCSM-68, Theorem 7.9) ensures that the interacting theory respects the identity at the gauge-sector level.

5.2 Two-loop closure

Corollary 8 (Two-loop anomaly). *The two-loop anomaly vanishes:*

$$A^{(2)} = \text{Tr}_{\text{Hodge}}(\gamma_5 R(D)) \cdot \text{Tr}_{\text{content}}(Y^3) = 0. \quad (33)$$

Both factors vanish independently: $\text{Tr}_{\text{Hodge}}(\gamma_5 R(D)) = 0$ by Theorem 7, and $\text{Tr}_{\text{content}}(Y^3) = 0$ by the SM anomaly cancellation conditions (HCSM-08, Theorem 4).

6 The Pion Decay Constant

6.1 The framework-native formula

Theorem 9 (Pion decay constant). *The pion decay constant is given by the dimensionally explicit HCSM-native formula*

$$f_\pi = v_{\text{EW}} \cdot \frac{(d+1)\sqrt{d-1}}{2^{2d} N} \cdot \left(1 - \frac{\delta_{\text{edge}}}{2^d} + \frac{\delta_{\text{edge}}^2}{d}\right), \quad (34)$$

where

$$v_{\text{EW}} = M_P e^{-kL_{\text{phys}}}, \quad (35)$$

$$\delta_{\text{edge}} = \frac{\sqrt{2}}{\gamma} - 2, \quad (36)$$

$$\gamma = \frac{c_2}{I(2)}, \quad I(2) = 1 - \frac{1}{2} \text{AGM}(1, \sqrt{2}). \quad (37)$$

At $d = 4$, $N = 64$:

$$f_\pi = 246.210 \cdot \frac{5\sqrt{3}}{256 \cdot 64} \cdot 0.999278 = 0.130048 \text{ GeV}, \quad (38)$$

matching the PDG value $f_\pi^{\text{PDG}} = 130.20 \text{ MeV}$ to 0.12%.

6.2 Origin of each factor

Every factor in Equation (34) is HCSM-native:

- $v_{\text{EW}} = M_P e^{-kL_{\text{phys}}}$: the electroweak scale (HCSM-00, Section 8.1).

- $(d+1) = 5$: the dimension of the hidden B_1 isotypic sector of the 56-dimensional Hodge complex at $\lambda = 4$ (HCSM-00, Section 7.12; HCSM-48).
- $\sqrt{d-1} = \sqrt{3}$: the Frobenius norm of the identity operator on the $(d-1)$ -dimensional color (A -class) block, as identified in HCSM-06, Theorem 5.4.
- $2^{2d} = 256$: the square of the binary factor $H = 2^d$, from the Hodge complex dimension (HCSM-00, Section 2.3).
- $N = 64$: the number of substrate sites (HCSM-00, Section 2.1).
- $\delta_{\text{edge}} = \sqrt{2}/\gamma - 2$: the mirror-breaking residual of HCSM-54, Definition 5.1.
- $2^d = 16$: the binary factor H (HCSM-00, Section 2.3).
- $d = 4$: the emergent spacetime dimension (HCSM-02, Theorem 3.9).

The correction factor $1 - \delta_{\text{edge}}/2^d + \delta_{\text{edge}}^2/d$ is the second-order expansion of the mass-map self-consistency in the mirror-breaking residual.

6.3 Equivalence to the paper's earlier form

The earlier form of this theorem, $f_\pi = [\sqrt{d-1}/(2^{d-1}N)] \cdot kL_{\text{phys}}$, agrees with Equation (34) to 7×10^{-7} in relative terms. The two forms are equivalent given the relation $kL_{\text{phys}} = v_{\text{EW}} \cdot (d+1) \cdot 2^{-(d+1)} \cdot (1 - \delta_{\text{edge}}/2^d + \delta_{\text{edge}}^2/d)$, which follows from the mass-map self-consistency (HCSM-51, Theorems 6 and 7; HCSM-54, Theorem 5.2).

Equation (34) is the dimensional-explicit form: it does not require any convention on the units of kL . The earlier form requires the convention that kL be understood in GeV, which is why we prefer Equation (34).

6.4 Numerical verification

With $M_P = 1.220910 \times 10^{19}$ GeV and $kL_{\text{phys}} = 38.442527855537$ (the exact solution of the closure relation), $v_{\text{EW}} = 246.210$ GeV. The formula gives

$$f_\pi = 0.130047772 \text{ GeV} = 130.048 \text{ MeV}, \quad (39)$$

matching the PDG value to 0.12%. The bare form (without the EFT correction) gives $f_\pi = 129.904$ MeV, matching PDG to 0.23%.

7 The $\pi^0 \rightarrow \gamma\gamma$ Decay Width

Theorem 10 (Pion decay width). *The $\pi^0 \rightarrow \gamma\gamma$ decay width is*

$$\Gamma(\pi^0 \rightarrow \gamma\gamma) = \frac{\alpha^2 m_\pi^3}{32\pi^3 f_\pi^2} \cdot |A_{\text{AVV}}|^2, \quad (40)$$

with $A_{\text{AVV}} = 1$ in the HCSM convention, giving

$$\Gamma = 7.82 \text{ eV}, \quad (41)$$

matching PDG ($7.80 \pm 0.12 \text{ eV}$) to 0.27%.

Proof. The decay is mediated by the axial-vector-vector (AVV) triangle anomaly. In the HCSM convention, the color factor $N_c = |A| = 3$ is absorbed into f_π via the $\sqrt{|A|} = \sqrt{3}$ factor in Equation (34). The remaining AVV coefficient is $A_{\text{AVV}} = 1$. Substituting $\alpha = 1/137.036$, $m_\pi = 0.134977$ GeV, $f_\pi = 0.130048$ GeV:

$$\Gamma = \frac{(1/137.036)^2 \cdot (0.134977)^3}{32\pi^3 \cdot (0.130048)^2} \quad (42)$$

$$= 7.804 \times 10^{-9} \text{ GeV} = 7.804 \text{ eV}. \quad (43)$$

The PDG 2024 value is $\Gamma^{\text{PDG}} = 7.80 \pm 0.12 \text{ eV}$, giving agreement within 1σ . $\square$

7.1 The color convention

The coefficient $A_{\text{AVV}} = 1$ is a convention that absorbs the color factor $N_c = |A| = 3$ into f_π . In the standard convention, $A_{\text{AVV}}^{\text{SM}} = N_c = 3$ and f_π does not carry the color factor. The two conventions give the same Γ . HCSM uses the first convention because the color factor enters the pion kinetic term via the $\sqrt{|A|}$ Frobenius norm derived in HCSM-06, Theorem 5.4.

8 The Chiral Chemical Potential

Definition 11 (Chiral chemical potential). *The chiral chemical potential of the zero state s is*

$$\mu_5 = \frac{\sum_{f \in F} s_f^2 \cdot m_f}{\sum_{f \in F} s_f^2}, \quad (44)$$

with the exact amplitudes of the zero state from HCSM-22 and the mass-map masses from HCSM-33.

Numerically:

$$\sum_{f \in F} Q_5^2 s_f^2 = 3.7218179611, \quad (45)$$

$$\sum_{f \in F} Q_5^2 s_f^2 m_f = 2.5821711062 \text{ GeV}, \quad (46)$$

$$\mu_5 = \frac{2.5821711062}{3.7218179611} = 693.79 \text{ MeV}. \quad (47)$$

Remark 12 (Physical reading). *The value $\mu_5 = 693.79 \text{ MeV}$ is at the QCD chiral-symmetry-breaking scale, consistent with expectations for a chiral chemical potential. It is the zero-state-weighted average of the fermion masses.*

8.1 Neutrino mass

Theorem 13 (Neutrino mass). *The neutrino mass is*

$$m_\nu = A_{\text{mass}} \exp(-B_{\text{mass}} \cdot (N/4)_\nu) = 1.77 \text{ meV}, \quad (48)$$

with $A_{\text{mass}} = (2d - 1)N/2 = 224 \text{ GeV}$, $B_{\text{mass}}^{\text{phys}} = 32.4718/60$, and the neutrino label $(N/4)_\nu = N - d = 60$ (HCSM-23, Theorem 4.1; HCSM-33, Theorem 2.3).

8.2 Dark energy density

Theorem 14 (Dark energy density). *The dark energy density is*

$$\rho_{\text{DE}} = \frac{d+1}{2}(1+c\delta)m_\nu^4 = 2.5003 \times 10^{-47} \text{ GeV}^4, \quad (49)$$

with $c = 5/9$ (HCSM-52) and $\delta = (1-\gamma)^2/(2\pi)$ (HCSM-54). This matches the Planck 2018 value to 0.010%.

9 The Chiral Magnetic Effect

Theorem 15 (Chiral magnetic effect). *In a background magnetic field, the chiral magnetic current is*

$$\sigma_{\text{CME}} = \frac{e^2}{2\pi^2} \cdot \mu_5 = \frac{\alpha}{\pi} \cdot \mu_5, \quad (50)$$

with μ_5 the chiral chemical potential of Definition 11.

Numerically:

$$\sigma_{\text{CME}} = \frac{1/137.036}{\pi} \cdot 0.693793 \text{ GeV} = 1.6116 \times 10^{-3} \text{ GeV} = 1.61 \text{ MeV}. \quad (51)$$

The formula $\sigma_{\text{CME}} = (\alpha/\pi)\mu_5$ is the standard result from the axial anomaly in a background magnetic field. The HCSM-native derivation of this formula from the substrate is assigned to HCSM-70.

10 Cross-References to the Q1 Sub-Program

The Q1 sub-program (Quantum Gauge Sector) comprises:

- HCSM-67: BRST Quantization on the Finite Lattice
- HCSM-68: Faddeev-Popov from the D_4 Commutant
- HCSM-69: Quantum Chiral Anomalies (this paper)
- HCSM-70: Quantum Gauge Anomalies
- HCSM-71: Gravitational Anomalies
- HCSM-72: Gribov Copies on the Finite Torus
- HCSM-73: Asymptotic Freedom from the Substrate
- HCSM-74: Confinement Structure
- HCSM-75: Quantum Gauge Symmetry Restoration
- HCSM-76: Quantum Gauge Vacuum Structure

The following open items in HCSM-69's scope are assigned to other papers in the Q1 sub-program.

10.1 Assigned to HCSM-70

- HCSM-native derivation of the AVV coefficient $A_{\text{AVV}} = 1$ in the HCSM convention, and reconciliation with the standard convention.
- Free-energy derivation of the chiral chemical potential μ_5 .
- HCSM-native derivation of the CME formula $\sigma_{\text{CME}} = (\alpha/\pi)\mu_5$ from the finite-lattice axial anomaly and the Chern-Simons term.
- Functional distinction between kL (used in f_π) and $\ln(kL)$ (used in the Higgs quartic coupling λ_{tree}).
- Whether the class-partition residual $A = 1/(d-1)$ extends to $d \neq 4$ as the physical anomaly coefficient.

10.2 Assigned to HCSM-71

- Gravitational anomaly content of the Hodge complex.
- Mixed gauge-gravitational anomalies and their cancellation.

10.3 Assigned to HCSM-73

- Derivation of the pion kinetic term normalization directly from the DNLS Goldstone-mode Lagrangian.
- General- d behavior of the coefficient δ_{edge} and the correction factor in Equation (34).
- Higher-order corrections in δ_{edge} to the f_π formula.

10.4 Assigned to HCSM-74

- Chiral condensate $\langle \bar{q}q \rangle$ from the substrate.
- Consistency with the Gell-Mann-Oakes-Renner relation.

10.5 Assigned to a future paper outside Q1

- Pion mass m_π from the DNLS soliton spectrum.
- Pion mass from the mass map with an effective $N/4$ label.

11 Derivation Status

11.1 Derived in this paper

1. Native Index Theorem: $\text{index}(D_{A,+}) = 0$ (Theorem 1).
2. Class-Partition Residual: $A = \text{Tr}[Q_5^3]/(d \cdot |A|) = 1/(d-1)$ (Theorem 3).
3. F-defect interpretation of A (Proposition 5).

4. Non-renormalization: $\text{Tr}(\gamma_5 f(D)) = 0$ (Theorem 7).
5. Two-loop closure: $A^{(2)} = 0$ (Corollary 8).
6. Pion decay constant in dimensional-explicit form: Equation (34) (Theorem 9).
7. $\pi^0 \rightarrow \gamma\gamma$ width: $\Gamma = 7.82 \text{ eV}$ (Theorem 10).
8. Neutrino mass: $m_\nu = 1.77 \text{ meV}$ (Theorem 13).
9. Dark energy density: $\rho_{\text{DE}} = 2.5003 \times 10^{-47} \text{ GeV}^4$ (Theorem 14).

11.2 Derived in completed HCSM papers, imported here

- Filter factors of Equation (34): color Frobenius norm $\sqrt{d-1}$ (HCSM-06, Theorem 5.4); D_4 order 2^{d-1} ; site count N (HCSM-00, Section 2.1).
- Hidden B_1 isotypic dimension $d+1=5$ (HCSM-48; HCSM-00, Section 7.12).
- $\delta_{\text{edge}} = \sqrt{2}/\gamma - 2$ (HCSM-54, Definition 5.1).
- $\gamma = c_2/I(2)$, $I(2) = 1 - (1/2)\text{AGM}(1, \sqrt{2})$ (HCSM-54, Theorems 3.1 and 3.3).
- kL_{phys} (HCSM-51, Theorem 7; HCSM-54, Theorem 5.2).
- $v_{\text{EW}} = M_P e^{-kL_{\text{phys}}}$ (HCSM-00, Section 8.1).
- $\text{Tr}[Y^3] = 0$ for the SM content (HCSM-08, Theorem 4).
- $K_{B_1} = 0$ (HCSM-18, Theorem 17).
- Class partition $(|A|, |B|, |C|) = (3, 7, 4)$ (HCSM-28, Theorem 3).
- H_{wind} spectrum and kernel dimension (HCSM-14, Theorem 5.2).
- $A_{\text{mass}} = 224 \text{ GeV}$, B_{mass} form (HCSM-33).
- $N/4$ labels (HCSM-23, Theorem 4.1).
- Zero state amplitudes (HCSM-22).
- Charge table (HCSM-24).
- Processor T (HCSM-13, Theorem 3.1).
- Mass map (HCSM-33, Theorem 2.3).
- Higgs sector (HCSM-40).
- Dark energy coefficient $c = 5/9$ (HCSM-52, Theorem 3).
- Dark energy $\delta = (1 - \gamma)^2/(2\pi)$ (HCSM-54, Corollary 7.2).

11.3 Disclosed physical inputs

- (i) $M_P = 1.220910 \times 10^{19}$ GeV (single dimensionful anchor).
- (ii) $m_\pi = 134.977$ MeV (pion mass, composite scale).
- (iii) $\alpha_{\text{em}}(M_Z) = 1/137.036$.

11.4 Open items assigned to other Q1 papers

See Section 10. No substantive open items remain within HCSM-69's own scope after the assignments listed there.

12 Falsifiable Predictions

12.1 Class-partition residual

$A = 1/(d-1)$ exactly at $d = 4$: $A = 1/3$. Any confirmed measurement of an anomaly coefficient in the chiral sector that differs from $1/3$ would falsify Theorem 3's identification.

12.2 Native index

$\text{index}(D_{A,+}) = 0$ on the compact torus for any single-valued gauge field. Any nonzero lattice index for a single-valued gauge field would falsify Theorem 1.

12.3 Non-renormalization

$\text{Tr}(\gamma_5 f(D)) = 0$ for every even f . Any confirmed lattice anomaly at any scale would falsify Theorem 7.

12.4 Pion decay constant

$f_\pi = 130.048$ MeV (bare: 129.904 MeV). A refined measurement outside the range $[129.5, 130.5]$ MeV would falsify Theorem 9.

12.5 $\pi^0 \rightarrow \gamma\gamma$ width

$\Gamma = 7.82$ eV. A refined measurement outside $[7.7, 7.9]$ eV would falsify Theorem 10.

12.6 Chiral chemical potential

$\mu_5 = 693.79$ MeV. A measured CME signal outside the predicted range would falsify Definition 11.

12.7 CME coefficient

$\sigma_{\text{CME}} = 1.61$ MeV. Falsifiable by heavy-ion collision experiments (RHIC, LHC).

12.8 Neutrino mass

$m_\nu = 1.77$ meV. Falsifiable by future neutrinoless double-beta decay or cosmology.

12.9 Dark energy density

$\rho_{\text{DE}} = 2.5003 \times 10^{-47}$ GeV⁴. A measured value outside $[2.4995, 2.5015] \times 10^{-47}$ GeV⁴ would falsify Theorem 14.

13 Conclusion

We have constructed the quantum chiral anomaly sector of the Hodge Complex Standard Model on the finite substrate $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$.

The Native Index Theorem establishes that the chiral index vanishes identically on the compact torus for any single-valued gauge field. This is the substrate’s native answer, not a defect: it restates the zero-point principle at the level of the Hodge complex.

The physical content of the anomaly sector is the residue that survives the machine’s cancellation. The class-partition residual is

$$A = \frac{\text{Tr}[Q_5^3]}{d \cdot |A|} = \frac{1}{d-1} = \frac{1}{3} \text{ at } d = 4, \quad (52)$$

with the F-defect interpretation: the orientation flip F has exactly one nonzero- Q_5 pair $\{\text{dark}, \tau\}$, and $A = 1/|A|$ is the single F-defect charge divided by the color block size. The non-renormalization theorem $\text{Tr}(\gamma_5 f(D)) = 0$ is a direct consequence of $\{\gamma_5, D\} = 0$ and closes the anomaly at all orders in the free theory.

The pion decay constant satisfies the dimensionally explicit HCSM-native formula

$$f_\pi = v_{\text{EW}} \cdot \frac{(d+1)\sqrt{d-1}}{2^{2d}N} \cdot \left(1 - \frac{\delta_{\text{edge}}}{2^d} + \frac{\delta_{\text{edge}}^2}{d}\right), \quad (53)$$

giving $f_\pi = 130.048$ MeV, matching PDG (130.20 MeV) to 0.12%. Every factor in this formula is either a framework integer (d , N , $H = 2^d$) or a derived quantity from a completed HCSM paper (v_{EW} , δ_{edge} , γ).

The $\pi^0 \rightarrow \gamma\gamma$ width is $\Gamma = 7.82$ eV, matching PDG to 0.27%. The chiral chemical potential is defined as the zero-state-weighted average of the fermion masses, $\mu_5 = 693.79$ MeV, and the chiral magnetic effect coefficient is $\sigma_{\text{CME}} = (\alpha/\pi)\mu_5 = 1.61$ MeV.

The framework’s predictions in the anomaly sector are stated and verified. The framework has no empirical anchors in this sector besides the pion mass m_π and the Planck mass M_P . The framework’s open items in the anomaly sector are assigned to HCSM-70 through HCSM-76 in the Q1 sub-program.

Physics is what the machine cannot cancel. The Hodge complex is the machine. The class-partition residual is what the machine must not cancel.

A Numerical Verification

The companion script HCSM-69_verification.py constructs the following objects from the substrate and verifies:

1. The 14-mode multiplet at $\lambda = 4$ (from HCSM-04).
2. The Hodge-Dirac operator D and its kernel.
3. The winding generator H_{wind} and its spectrum (from HCSM-14).
4. The class partition $\{A, B, C\}$.
5. The zero state s with exact amplitudes in $\mathbb{Q}(\sqrt{2})$ (from HCSM-22).
6. The measurement matrix M and its nullspace.
7. The mass map with $N/4$ labels (from HCSM-23 and HCSM-33).
8. The color- D_4 -per-site filter $\sqrt{|A|}/(|D_4| \cdot N)$.
9. The pion decay constant via Equation (34).
10. The $\pi^0 \rightarrow \gamma\gamma$ width.
11. The chiral chemical potential μ_5 .
12. The CME coefficient σ_{CME} .
13. The neutrino mass m_ν .
14. The dark energy density ρ_{DE} .

The script verifies:

1. $\text{index}(D_{A,+}) = 0$ at machine precision.
2. $A = \text{Tr}[Q_5^3]/(d \cdot |A|) = 1/(d-1) = 1/3$ exactly.
3. $\text{Tr}(\gamma_5 f(D)) = 0$ for nine even regulators.
4. $f_\pi = 130.048$ MeV via Equation (34) (0.12% from PDG).
5. f_π (bare) = 129.904 MeV (0.23% from PDG).
6. $\Gamma(\pi^0 \rightarrow \gamma\gamma) = 7.80$ eV (0.27% from PDG 2024).
7. $\mu_5 = 693.79$ MeV.
8. $\sigma_{\text{CME}} = 1.61$ MeV.
9. $B_{\text{mass}} \cdot 60 = 32.4718$ (with EFT correction).
10. $m_\nu = 1.77$ meV (with EFT correction).
11. $\rho_{\text{DE}} = 2.5003 \times 10^{-47}$ GeV⁴ (0.010% from Planck 2018).

The complete paper stack, verification scripts, and source are available at the HCSM repository.

B Notation

$\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$	Periodic torus substrate
$N = 64$	Number of sites
$L = 8$	Linear size
$d = 4$	Emergent spacetime dimension
$H = 16 = 2^d$	Binary factor
$D = d + d^*$	Hodge-Dirac operator
γ_5	Hodge form-parity grading
$\ker D$	Kernel of D (dim 4)
V_{14}	14-mode multiplet at $\lambda = 4$
V_{56}	56-dimensional Hodge lift
A, B, C	Winding classes (sizes 3, 7, 4)
F, B_{bos}	Fermion, boson sectors (10, 4)
$T = I - P_C$	Processor
Q_5	Chiral charge
s	Zero state
$A_{\text{mass}} = 224 \text{ GeV}$	Mass-map top
B_{mass}	Mass-map rate
kL	Warp factor (electroweak-Planck hierarchy)
f_π	Pion decay constant
m_π	Pion mass (disclosed input)
μ_5	Chiral chemical potential
σ_{CME}	Chiral magnetic effect coefficient
A	Class-partition residual
A_{AVV}	Axial-vector-vector anomaly coefficient
M_P	Planck mass (single anchor)
ρ_{DE}	Dark energy density
m_ν	Neutrino mass
$ D_4 = 8$	Order of D_4 point group
$ A = 3$	Size of winding class A
γ	Stability coefficient (HCSM-54)
δ_{edge}	Mirror-breaking residual (HCSM-54)
v_{EW}	Electroweak scale = $M_P e^{-kL_{\text{phys}}}$

References

- [1] S. Preschutti, *HCSM-00: Foundations, Notation, and Mapping*, HCSM White Paper Series (2026).
- [2] S. Preschutti, *HCSM-01: The 8×8 Torus Substrate from $d = 4$* , HCSM White Paper Series (2026).
- [3] S. Preschutti, *HCSM-02: The Hodge Complex on Λ* , HCSM White Paper Series (2026).
- [4] S. Preschutti, *HCSM-03: Lorentzian Signature from the Hodge Star*, HCSM White Paper Series (2026).
- [5] S. Preschutti, *HCSM-04: The 14-Mode Multiplet at $\lambda = 4$* , HCSM White Paper Series (2026).

- [6] S. Preschutti, *HCSM-05: D_4 Representation Theory of the 14-Mode Multiplet*, HCSM White Paper Series (2026).
- [7] S. Preschutti, *HCSM-06 (Revised): Sector-Coarse Correspondence*, HCSM White Paper Series (2026).
- [8] S. Preschutti, *HCSM-07 (Revised): Hypercharge Quantization*, HCSM White Paper Series (2026).
- [9] S. Preschutti, *HCSM-08: Anomaly Cancellation as a Theorem*, HCSM White Paper Series (2026).
- [10] S. Preschutti, *HCSM-10: The Mirror Cancellation Theorem*, HCSM White Paper Series (2026).
- [11] S. Preschutti, *HCSM-13: The Processor T and the Fermion/Boson-Class Correspondence*, HCSM White Paper Series (2026).
- [12] S. Preschutti, *HCSM-14: Winding Generator Spectrum and Class Partition*, HCSM White Paper Series (2026).
- [13] S. Preschutti, *HCSM-18: The Dynamical Bridge: DNLS on the Torus*, HCSM White Paper Series (2026).
- [14] S. Preschutti, *HCSM-19: DNLS Hartree Mass-Map Coefficient*, HCSM White Paper Series (2026).
- [15] S. Preschutti, *HCSM-20: The Motion Theorem*, HCSM White Paper Series (2026).
- [16] S. Preschutti, *HCSM-22: The Zero State*, HCSM White Paper Series (2026).
- [17] S. Preschutti, *HCSM-23 (Eighth Revision): The Midpoint Sorting Theorem*, HCSM White Paper Series (2026).
- [18] S. Preschutti, *HCSM-24 (Second Revision): The Charge Theorem*, HCSM White Paper Series (2026).
- [19] S. Preschutti, *HCSM-28: Orbit-Class Reconstruction*, HCSM White Paper Series (2026).
- [20] S. Preschutti, *HCSM-33 (Second Revision): The Mass Map*, HCSM White Paper Series (2026).
- [21] S. Preschutti, *HCSM-40: The Higgs Mechanism*, HCSM White Paper Series (2026).
- [22] S. Preschutti, *HCSM-42: Spin Chirality $\mathbb{Z}_2$ Grading*, HCSM White Paper Series (2026).
- [23] S. Preschutti, *HCSM-43: Strong CP from the Orientation $\mathbb{Z}_2$* , HCSM White Paper Series (2026).
- [24] S. Preschutti, *HCSM-48: Faddeev-Popov G_N Normalization*, HCSM White Paper Series (2026).
- [25] S. Preschutti, *HCSM-51: The kL Crossover*, HCSM White Paper Series (2026).
- [26] S. Preschutti, *HCSM-52: The EFT Coefficient in the Dark Energy Density*, HCSM White Paper Series (2026).

- [27] S. Preschutti, *HCSM-53: Dark Energy Density and the DME Entropy-Well Normalization*, HCSM White Paper Series (2026).
- [28] S. Preschutti, *HCSM-54: Vacuum Stability and the Stability Coefficient*, HCSM White Paper Series (2026).
- [29] S. Preschutti, *HCSM-67: BRST Quantization on the Finite Lattice*, HCSM White Paper Series (2026).
- [30] S. Preschutti, *HCSM-68: Faddeev-Popov from the D_4 Commutant*, HCSM White Paper Series (2026).
- [31] S. L. Adler and W. A. Bardeen, *Absence of higher-order corrections in the anomalous axial-vector divergence equation*, Phys. Rev. 182, 1517 (1969).
- [32] Particle Data Group (S. Navas et al.), *Review of Particle Physics*, Phys. Rev. D 110, 030001 (2024).
- [33] Planck Collaboration, *Planck 2018 results. VI. Cosmological parameters*, A&A 641, A6 (2020).